$\text{Statistics}_{a}ssignment_{I}MAPP$

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November 2026

1 Introduction to the problem: Ehrenfest's dogs

Consider a system of two dogs sharing a total of N fleas. If the first dog carries N_+ fleas, the second dog carries $N_- = N - N_+$ fleas. The fleas are free to jump from one dog to the other, one flea per step. Each flea can therefore be treated as a two-state variable that flips between the values $+1$ and -1: a value of $+1$ means the flea is on the first dog, and a value of -1 means it is on the second dog.

To describe the state of the whole system at a given step, we track the *asymmetry*

$$A = \frac{N_+ - N_-}{N}, \qquad A \in [-1, 1]. \tag{1}$$

The asymmetry is $A = 0$ when both dogs carry the same number of fleas (that is, when the number of $+1$ states equals the number of -1 states); we call this the **equilibrium state**. At the extremes, $A = +1$ when all the fleas are on the first dog and $A = -1$ when all the fleas are on the second dog.

The total number of fleas (states), N, can be regarded as the dimension of the system. Since each jump corresponds to a random flip of a single state between $+1$ and -1, the system at equilibrium is described by a binomial distribution.

Initially, all the fleas are on the first dog, so $A = +1$ at the first step.

Throughout this exercise, our task is to track the evolution of A as the states are allowed to flip freely, for systems of different dimension N.

1.1 Theory and background

This setup is the Ehrenfest urn model, introduced in 1907 to show how irreversible behaviour can emerge from reversible microscopic dynamics. At each step one flea is chosen at random and moved to the other dog, so a flea leaves the first dog with probability N_+/N. On average, this gives

$$\langle A_{t+1} \rangle = (1 - 2N) \langle A_t \rangle \quad \Rightarrow \quad \langle A_t \rangle = (1 - 2N)^t \approx e^{-2t/N}, \tag{2}$$

so the asymmetry relaxes exponentially towards zero on a timescale of order N.

At equilibrium, each flea is independently equally likely to be on either dog, so $A = \frac{1}{N} \sum_{i=1}^{N} s_i$ is the mean of N independent variables $s_i = \pm 1$, and

N_+ follows a binomial distribution, $P(N_+) = NN_+2^{-N}$. By the central limit theorem, for large N the asymmetry is approximately Gaussian,

$$A \sim \mathcal{N}(0, 1N)\,, \qquad \sigma_A = \frac{1}{\sqrt{N}}, \tag{3}$$

so the fluctuations about equilibrium shrink as N grows.

2 Simulation and Analysis

This section will describe step by step instructions for simulation and analyses of the problem.

2.1 Simulation instructions

1. Define global variables such as the seed for randomness, a list of even dimensions (N), total number of runs, tolerance for maximum number of flips allowed and maximum allowed number of entries for histogram.
2. Define the initial state initializer function where all the fleas are in state +1 by creating a list of all states and also define a function that randomly flips one state from +1 to -1 or reverse at one step depicting random flea jumping.
3. Define a run function for a given even dimension, initialize the initial state using the defined function, and define observables, namely, the number of flips for a round trip (system returning to the initial state), number of flips before first visit to equilibrium, number of flips before last visit to equilibrium, number of visit to equilibrium in a round trip.
4. Inside the function, run a loop till the system completes a round trip or reaches the tolerance for flips. Inside the loop you keep track of the values of A and the number of flips and update the observables according to the value of A (asymmetry) in the system. Make sure to exit the loop on completing the round trip.
5. Define another observable within the function to keep track of average time between two equilibrium states:

$$\Delta_{eq} = \frac{t_{\text{last}} - t_{\text{first}}}{n_{eq} - 1} \tag{4}$$

2.2 Analyses instructions

1. Define a function to plot observable average as a function of the system dimension and also fits a 2 parameter curve onto the plot. The function can take parameters like, dimension list, observable per dimension, error on observable value for error plot, function for curve fit, initial values of the two parameters. Other than plotting, this functions can also return parameter values from curve fit and a 2x2 covariance matrix for the respective parameters.
2. Define the Poisson interval (à la Garwood). For an observed count y,

$$\mu_{\text{lo}} = 12\,\chi^2_{\alpha}(2y), \qquad \mu_{\text{hi}} = 12\,\chi^2_{1-\alpha}\big(2(y+1)\big) \tag{5}$$

Here $\chi^2_p(k)$ is the p-th quantile of a χ^2 distribution with k degrees of freedom. We take $\alpha \approx 0.1587$, so that the interval covers 68.27% (i.e. 1σ). When $y = 0$ the lower limit is just $\mu_{\rm lo} = 0$. The upper and lower errors, and their average, are

$$\delta_+ = \mu_{\rm hi} - y, \qquad \delta_- = y - \mu_{\rm lo}, \qquad \delta = \frac{\delta_+ + \delta_-}{2} \tag{6}$$

We use δ as the error on y. For small counts, this works better than the usual $\sqrt{y}$. (6)

3. Define a function to plot the histogram for count vs asymmetry values in a roundtrip for a given dimension. This function should also fit a gaussian probability density function onto the histogram with estimated parameters of interests, which are, mean, standard deviation and normalizing constant. The input parameters to this function can be the data and dimension, and it returns along with the plots, also the values of the estimated parameters, their covariance matrix and the total area of the histogram since for a good fit it should be equal to the area under the gaussian curve.

$$f(A) = \frac{I}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(A-\mu)^2}{2\sigma^2}\right] \tag{7}$$

4. Define a function for producing histogram for values of observable across total number of runs (the global variable defined earlier). The input parameters of this function can be the data and dimension, and it returns a hist plot along with sample mean and error.
5. Define a function that plots the error ellipse for the fitted parameters $\hat{a}$ and $\hat{b}$. To do this, we can diagonalise the covariance matrix C:

$$C\,\vec{v}_i = \lambda_i\,\vec{v}_i, \qquad i = 1, 2 \tag{8}$$

the eigenvalues $\lambda_{1,2}$ set the size of the semi-major and semi-minor axes, and the eigenvectors give its tilt (eigenvectors point in the direction of maximum and minimum variance respectively). The semi-axes are $s\sqrt{\lambda_{1,2}}$, with

$$s = \sqrt{\chi^2_{\rm CL}(2)} = \sqrt{-2\ln(1 - {\rm CL})} \tag{9}$$

i.e. the square root of the χ^2 quantile for the chosen confidence level. We use 2 degrees of freedom since there are two parameters.
6. Finally, for each dimensionality, create mean and error zeros list for three observables namely, number of flips before the system: returns to inital state, reaches first equilibrium, average time between equilibrium. Loop inside the dimension list, where you start round trips for each dimension "total number of run" times (global variable) and collect data for each round trip by creating an array for all the runs. Loop for total number of runs and create a list that collects all the A (asymmetry) values after a global tolerance variable check. Within the dimension list loop, run the asymmetric and observable histogram

functions defined earlier. Print the results for each observable (averages and asymmetry mean, rms and norm) Collect means and corresponding errors as a function of the dimension N in the zeros list initiated earlier in this step.
7. Study the different fit plots for all four observables (round trip flips, flips to first equilibrium, average time between two equilibrium states and asymmetry) against dimensionality of the system and for the fit plots, study also the error ellipse for the parameters and draw conclusions from all the plots.

2.3 Suggested python tips and checks